

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K₄ Fe (CN)₆. After filtering, the precipitate is dried and weighed. The reactions which occur are $\text{ZnS} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$ $2\text{ZnCl}_2 + \text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Zn}_2\text{Fe}(\text{CN})_6 + 4\text{KCl}$ If a 2 gram sample of rock yields 0.969 gram of Zn₂Fe(CN)₆, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn₂ Fe (CN)₆ = 342.6.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

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- Mass of Zn₂ Fe (CN)₆ = 0 . 9 6 9 g .
- Molecular weight of Zn₂ Fe (CN)₆ = 3 4 2 . 6 .
- Atomic weight of Zn = 6 5 . 4 .
- We need to find the mass of Zn in the 0 . 9 6 9 g of Zn₂ Fe (CN)₆ .
- Mass of Zn in Zn₂ Fe (CN)₆ = (2 * Atomic weight of Zn) / Molecular weight of Zn₂ Fe (CN)₆ * Mass of Zn₂ Fe (CN)₆ .
- Mass of Zn = (2 * 6 5 . 4) / 3 4 2 . 6 * 0 . 9 6 9 = 0 . 3 7 3 g .
- Percentage of Zn in the sample = (Mass of Zn / Mass of rock sample) * 1 0 0 = (0 . 3 7 3 / 2) * 1 0 0 = 1 8 . 6 5 % âĒĒ 1 8 % Zn .

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< answer >

F . 1 8 % Zn

</answer>